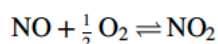


PROBLEM SET 5

(Due Oct 7, 2022 @ 12:00 pm in Gradescope and WileyPlus)

This assignment is to be completed individually. The number in parentheses next to each problem shows how many points the problem is worth in the overall assignment. Please show all work and use correct significant figures to receive full credit. Be sure to follow the problem formatting instructions to avoid unnecessary deductions. If you use an *equation-solving tool* (calculator, APEx, Solver, MATLAB), write out the equation(s) and note the tool that you used; otherwise, show all hand calculations used to solve any equations.

- (20 pts)** Liquid methanol is fed to a space heater at a rate of 12.0 L/h and burned with excess air. The product gas is analyzed and the following dry-basis mole percentages are determined: $\text{CH}_3\text{OH} = 0.45\%$, $\text{CO}_2 = 9.03\%$, and $\text{CO} = 1.81\%$.
 - Draw and label a flowchart and verify that the system has zero degrees of freedom.
 - Calculate the fractional conversion of methanol, the percentage excess air fed, and the mole fraction of water in the product gas.
 - Suppose the combustion products are released directly into a room. What potential problems do you see and what remedies can you suggest?
- (5 pts)** Use the ideal-gas equation of state to estimate the molar volume in m^3/mol and the density of air in kg/m^3 at 40°C and a gauge pressure of 3.0 atm.
- (15 pts)** An ideal-gas mixture contains 35% helium, 20% methane, and 45% nitrogen by volume at 2.00 atm absolute and 90°C . Calculate (a) the partial pressure of each component, (b) the mass fraction of methane, (c) the average molecular weight of the gas, and (d) the density of the gas in kg/m^3 .
- (30 pts)** The oxidation of nitric oxide



takes place in an isothermal batch reactor. The reactor is charged with a mixture containing 20.0 volume percent NO and the balance air at an initial pressure of 380 kPa (absolute).

- Assuming ideal-gas behavior, determine the composition of the mixture (component mole fractions) and the final pressure (kPa) if the conversion of NO is 90%.
- Suppose the pressure in the reactor eventually equilibrates (levels out) at 360 kPa. What is the equilibrium percent conversion of NO? Calculate the reaction equilibrium constant at the prevailing temperature, $K_p[(\text{atm})^{-0.5}]$, defined as

$$K_p = \frac{(p_{\text{NO}_2})}{(p_{\text{NO}})(p_{\text{O}_2})^{0.5}}$$

where $p_i(\text{atm})$ is the partial pressure of species i (NO_2 , NO , O_2) at equilibrium.

- Assuming that K_p depends only on temperature, estimate the final pressure and composition in the reactor if the feed ratio of NO to O_2 and the initial pressure are the same as in Part (a), but the feed to the reactor is pure O_2 instead of air.
- Replace the partial pressures in the expression for K_p , and use the result to explain how reactor pressure influences the conversion of NO to NO_2 .

(30 pts) The remainder of the assignment (3 problems) will be completed individually online using WileyPLUS. *You do not have to turn in any paperwork with this portion of the assignment.* Use the link for your class on the Moodle site, and then you can access the Assignment within WileyPLUS. The due date for the WileyPLUS completion is the same as for the homework assignment – the beginning of your class period. Note that the WileyPLUS assignment cannot be submitted late.

Honors Program Problem

A stream of air enters a 7.50-cm ID pipe with a velocity of 60.0 m/s at 27°C and 1.80 bar (gauge). At a point downstream, the air flows through a 5.00 cm ID pipe at 60°C and 1.53 bar (gauge). What is the average velocity of the gas at this point?